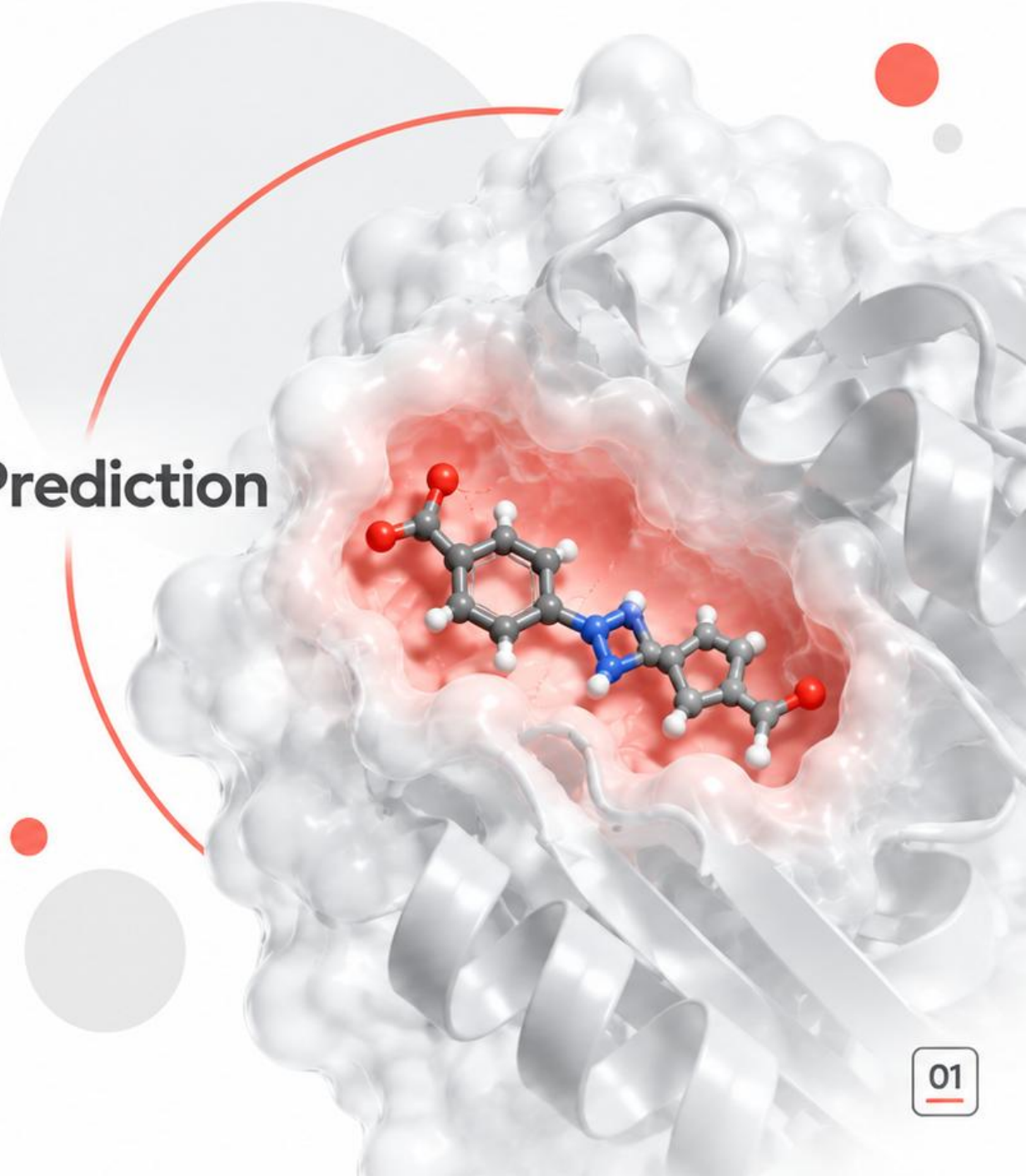
Molecular Docking:

From Protein Structure to Binding Prediction

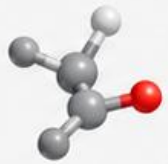
Pertemuan 1 — Preparasi dan Validasi Docking

Molecular docking helps predict how a ligand binds to a target protein.



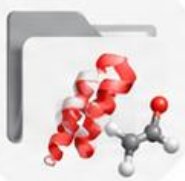
Learning goals

Tujuan Pembelajaran



01

menjelaskan prinsip **molecular docking**



02

memilih struktur **reseptor dan ligan**



03

melakukan preparasi **protein dan ligan**



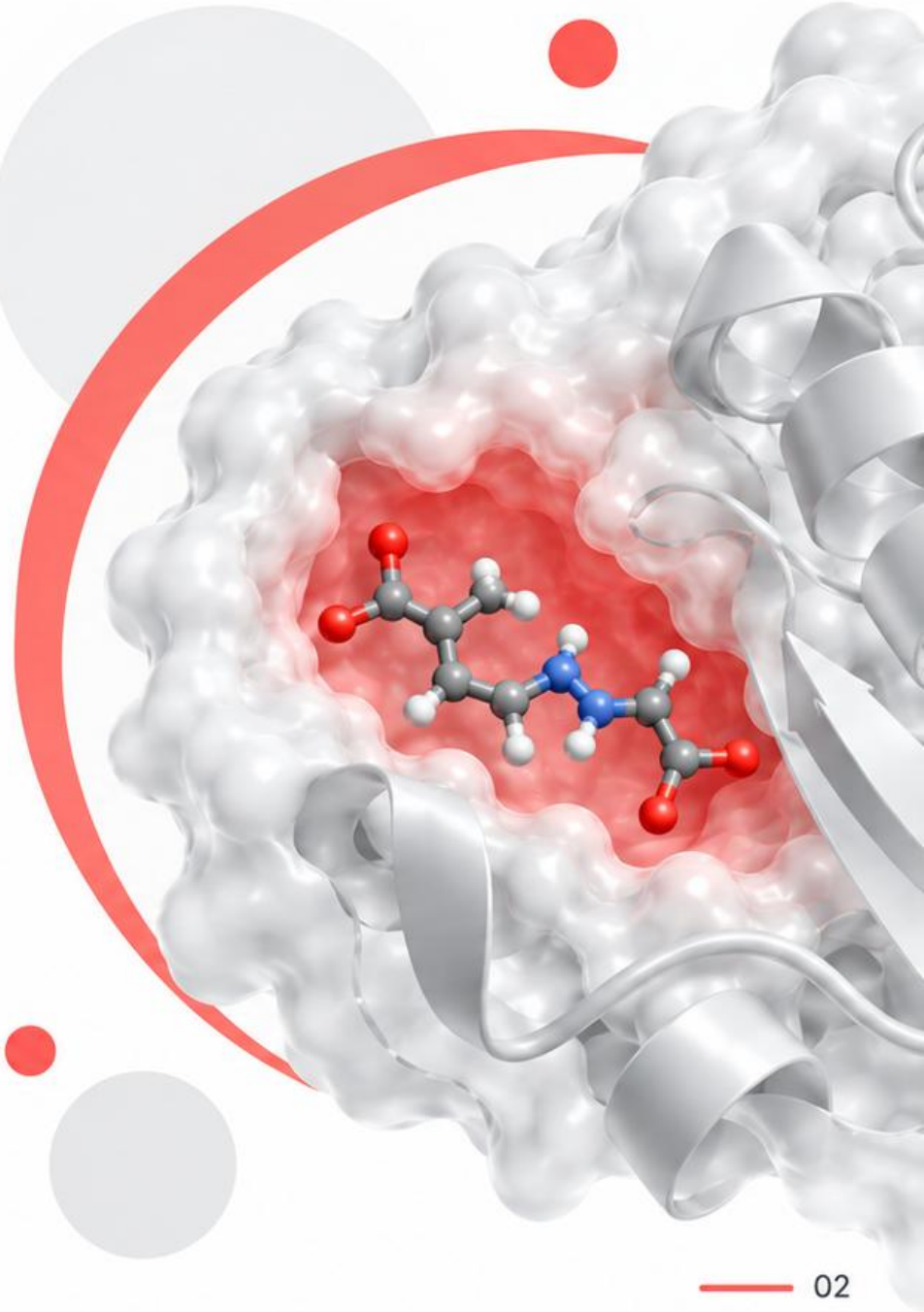
04

menentukan **binding site / grid box**



05

memahami validasi metode dengan **redocking**



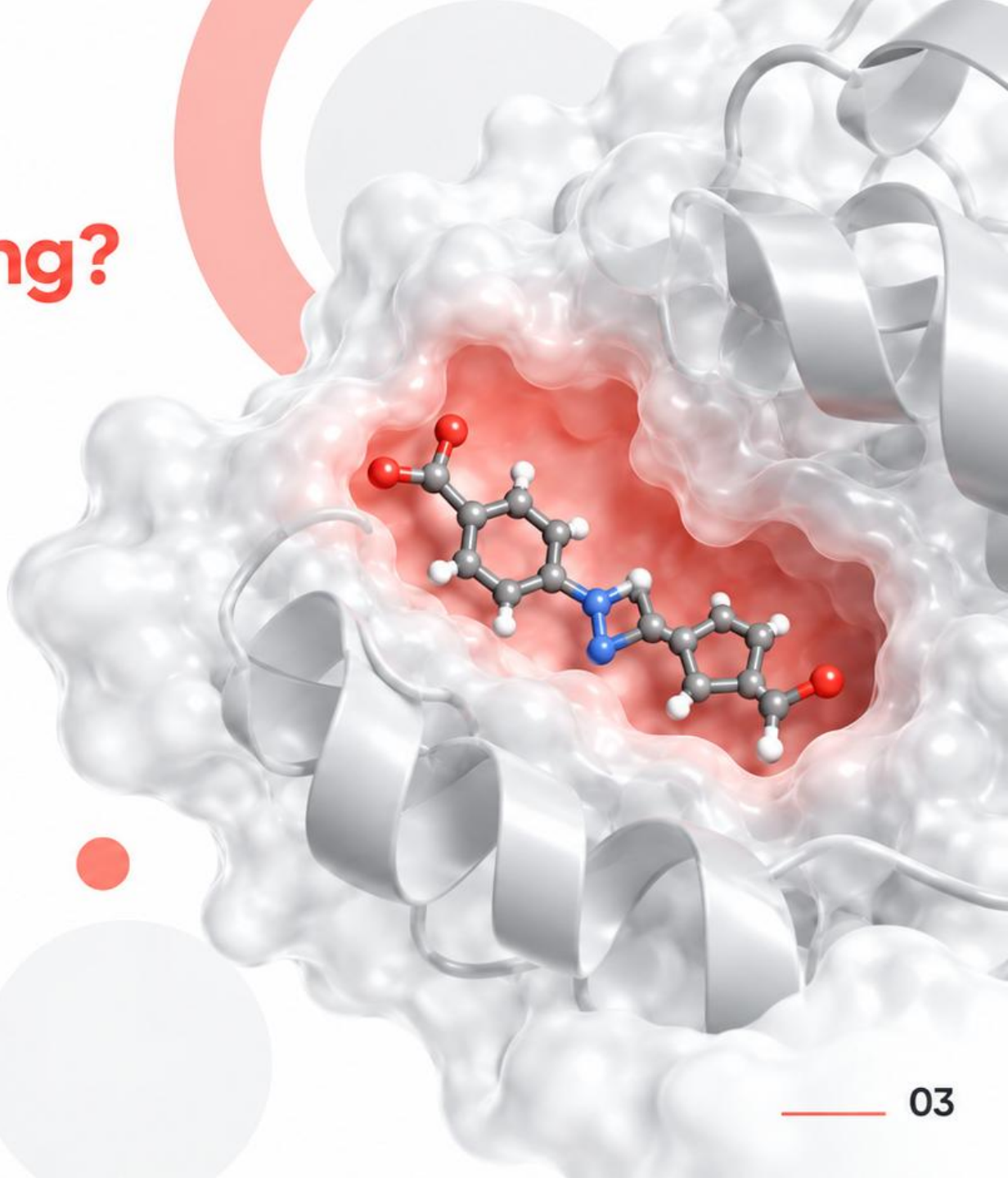
Mengapa Molecular Docking Penting?

apakah ligan dapat masuk ke **binding site**

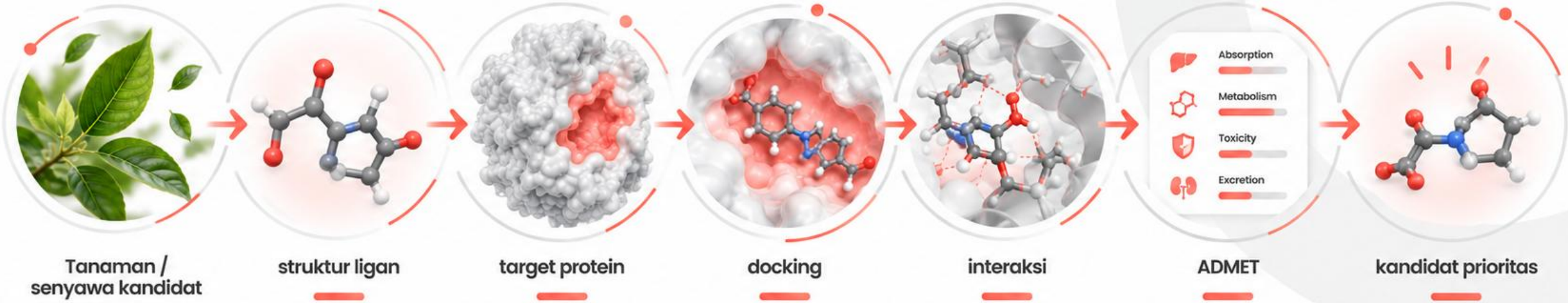
pose ikatan yang paling mungkin

estimasi **afinitas** ikatan

jenis **interaksi** ligan–reseptor



Posisi Docking
dalam Drug Discovery



Tanaman / senyawa kandidat

struktur ligan

target protein

docking

interaksi

ADMET

kandidat prioritas

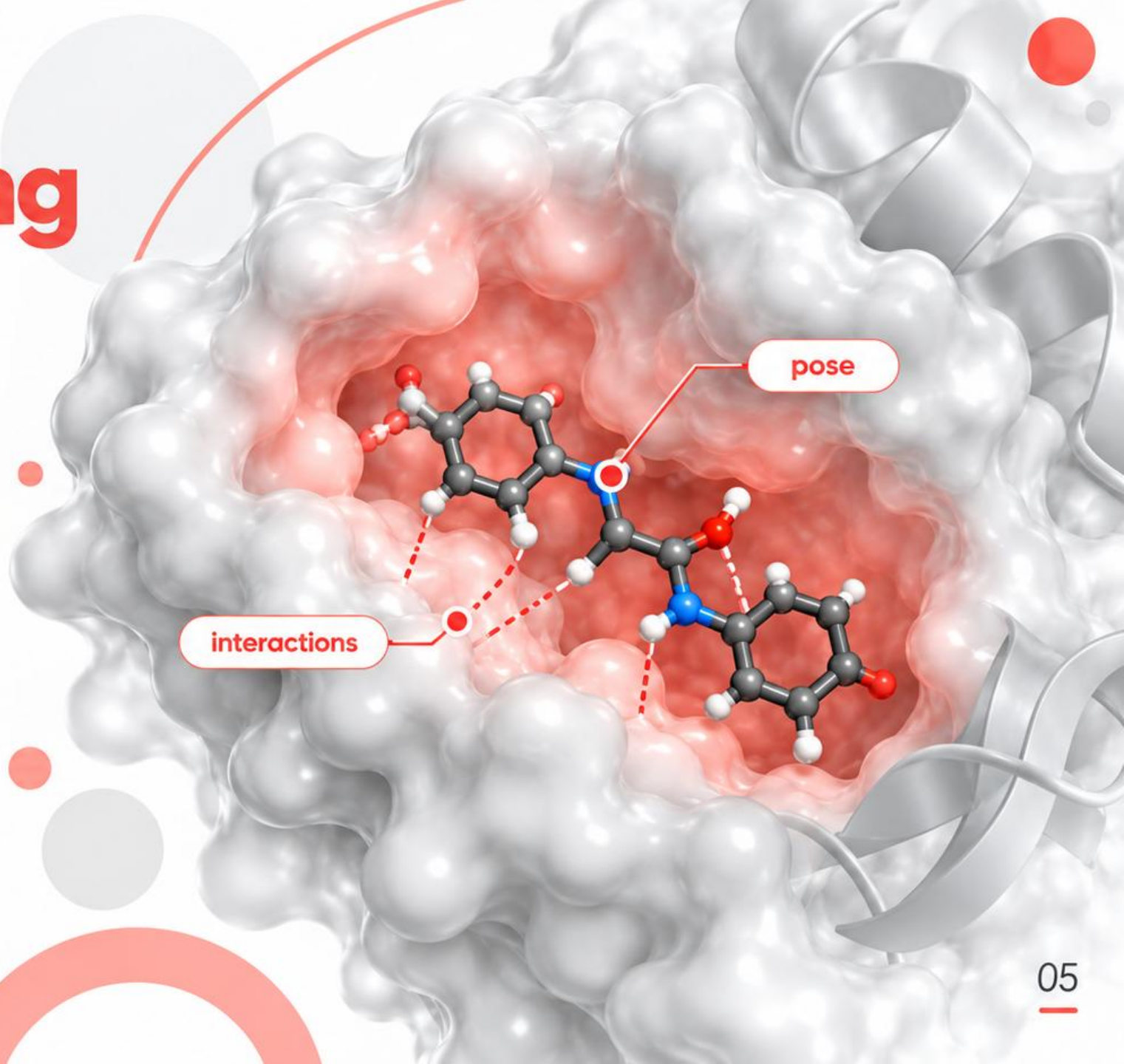
Output Docking

binding affinity
kcal/mol

pose ligan

interaksi:
H-bond, hidrofobik,
van der Waals, π-interaction

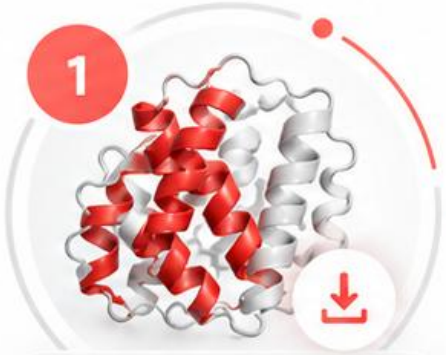
residu asam amino
penting



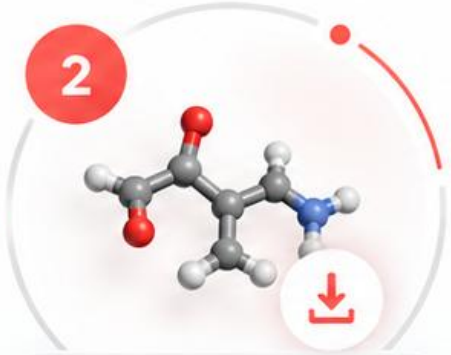
Source: AutoDock Vina

Workflow Praktik Kelas

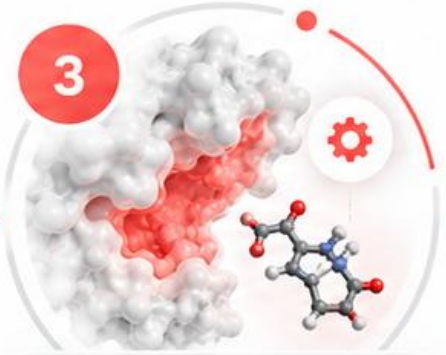
Alur umum praktikum molecular docking



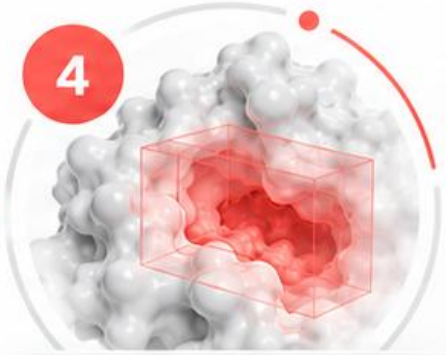
1 Download protein reseptor



2 Download ligan



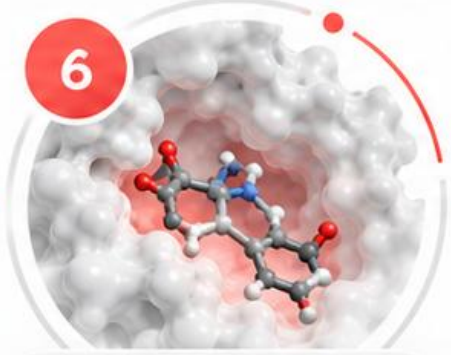
3 Preparasi protein dan ligan



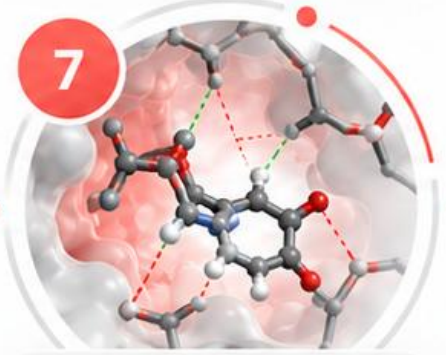
4 Prediksi / penentuan binding site



5 Redocking ligan ko-kristal



6 Docking ligan kandidat



7 Visualisasi interaksi

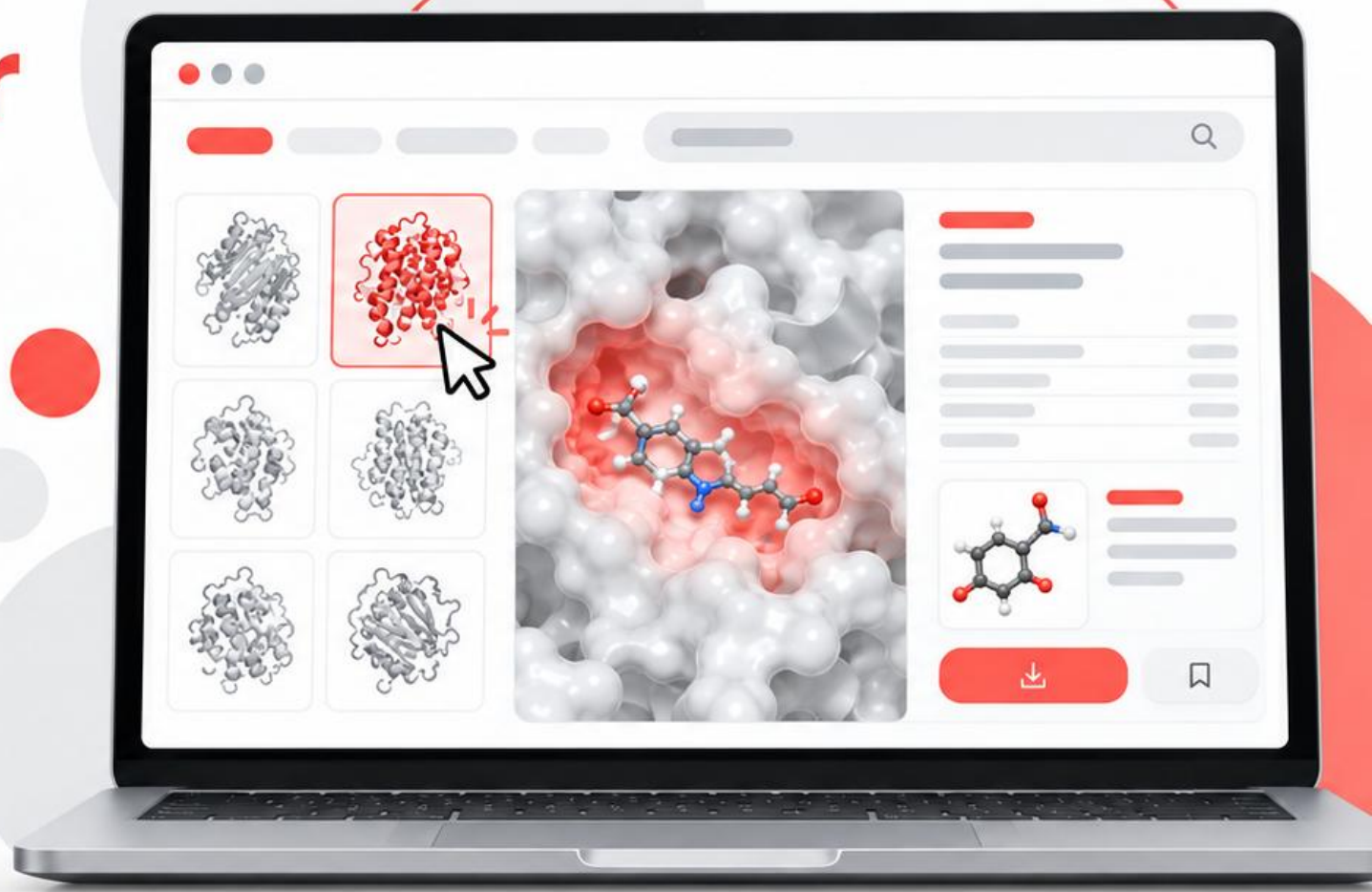


8 ADMET & drug-likeness

Protein: RCSB/UniProt • Ligan: PubChem • Site: PrankWeb/CavityPlus

Memilih Protein Reseptor

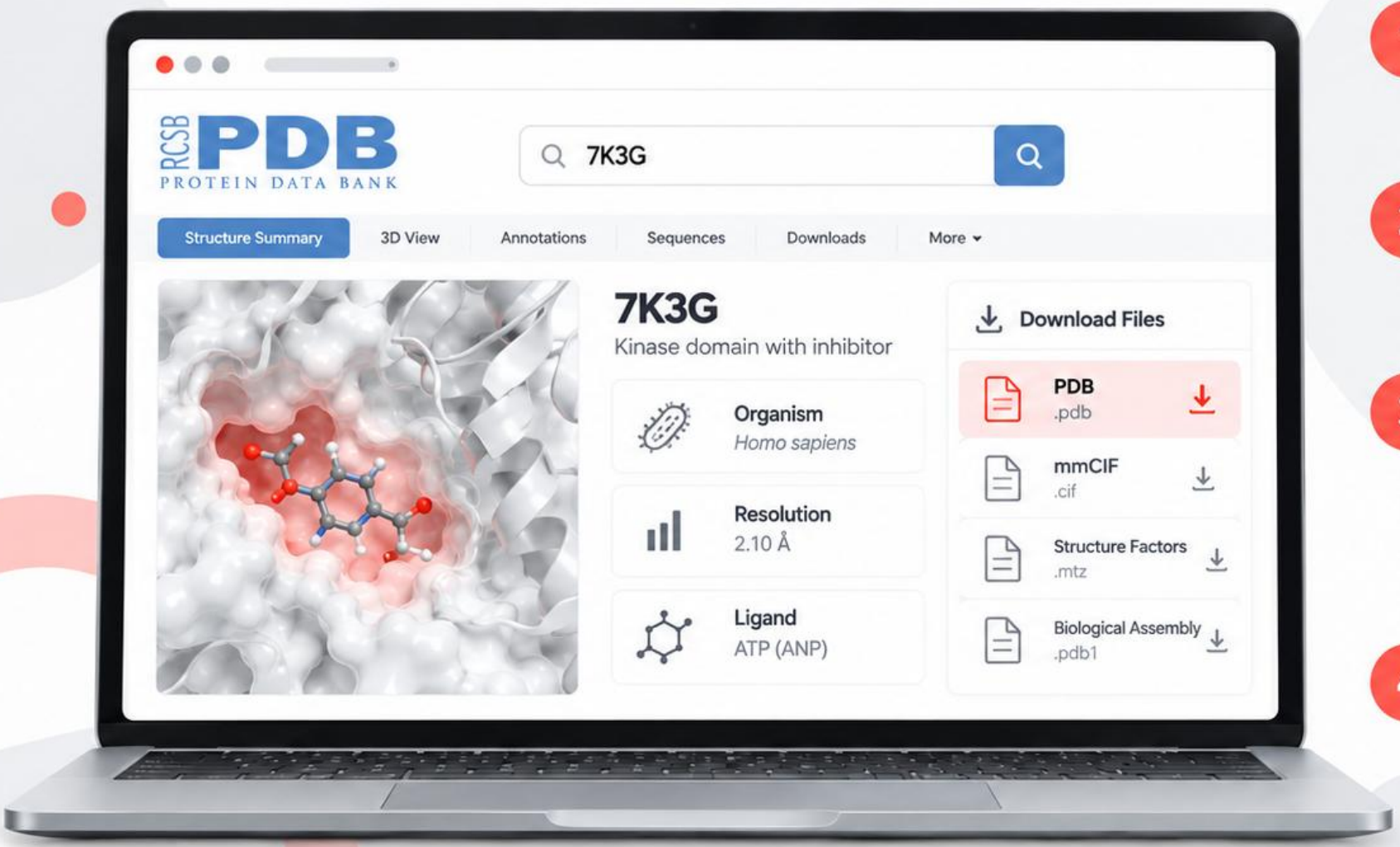
- spesies sesuai (ideal: *Homo sapiens*)
- resolusi baik (< 3 Å)
- ada ligan ko-kristal
- binding site jelas
- hindari *missing residues* penting



Prefer structure with biologically relevant binding information.

Download Struktur Protein

Demonstrasi dari RCSB PDB



1. **buka** RCSB PDB



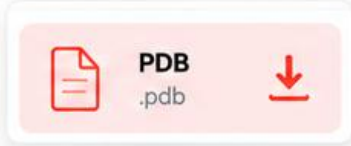
2. **cari** PDB ID



3. **cek** organisme, resolusi, dan ligan



4. **download** file PDB



Download Ligan dari PubChem

1. cari nama ligan / senyawa

Misalnya: aspirin, glutamate, atau nama IUPAC yang kamu inginkan.

2. buka PubChem

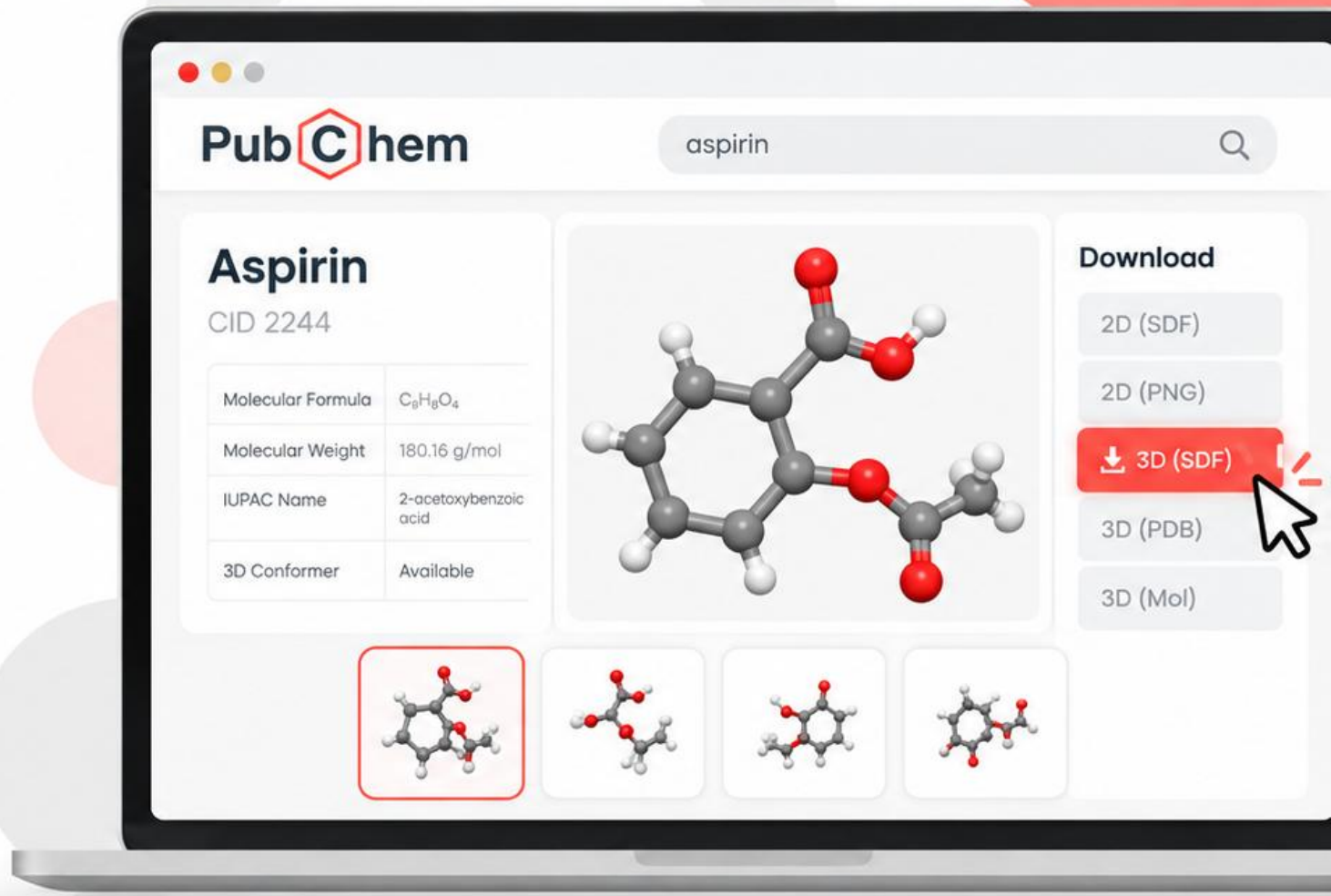
Kunjungi https://pubchem.ncbi.nlm.nih.gov

3. download 3D conformer

Pada halaman senyawa, pilih format 3D dan unduh strukturnya.

4. format: SDF

Simpan struktur ligan dalam format SDF (Structure Data File).



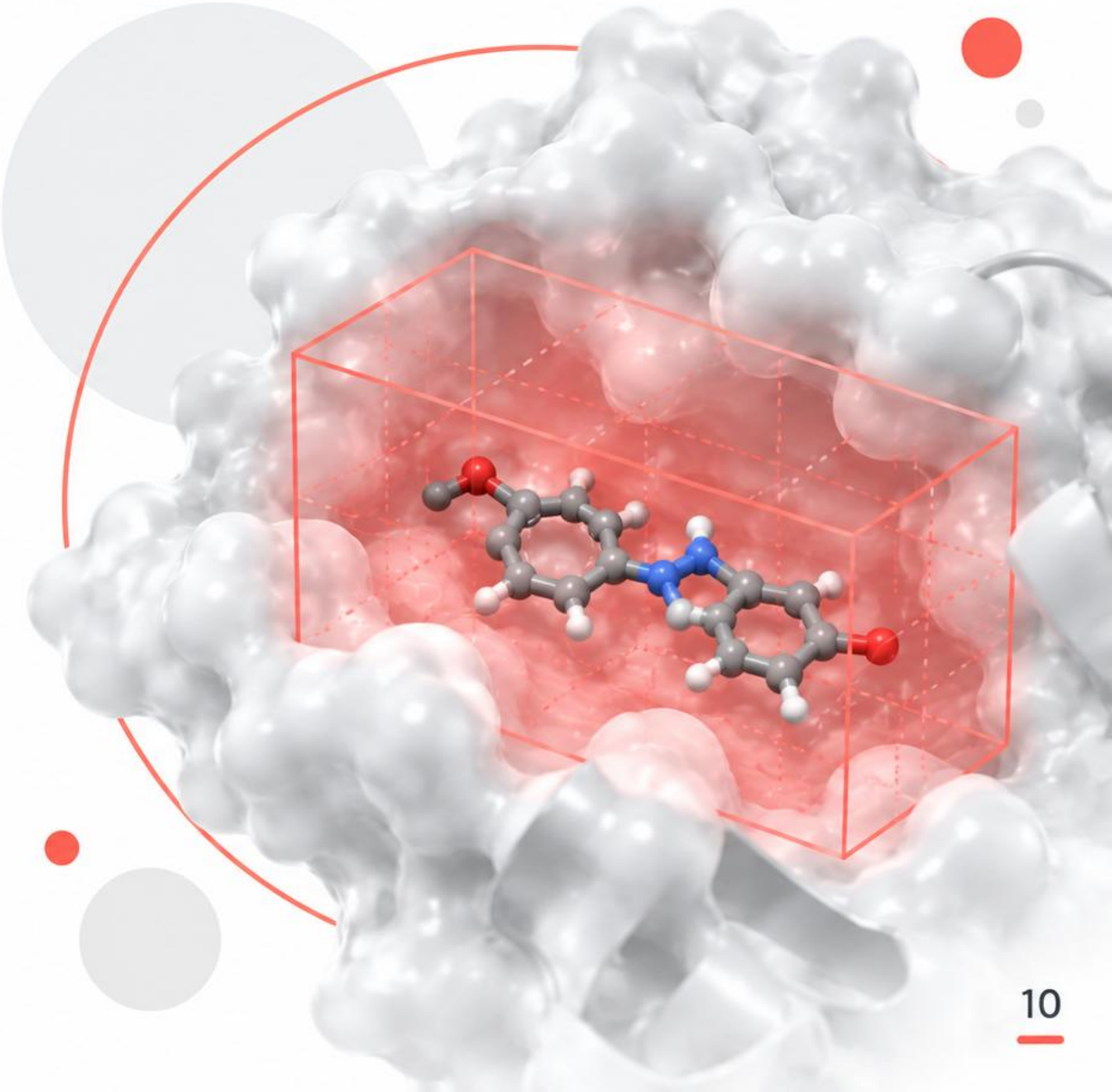
Source: PubChem Structures

Binding Site dan Grid Box

Grid box adalah ruang pencarian tempat ligan dicoba masuk.

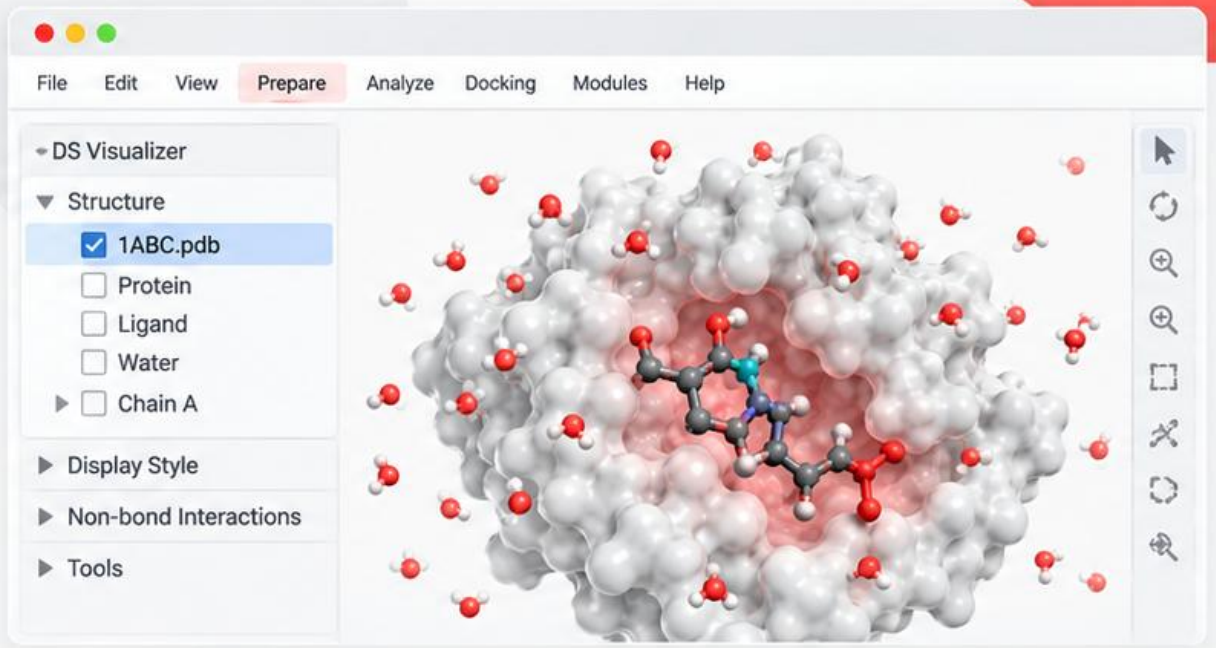
Grid box dapat ditentukan dari:

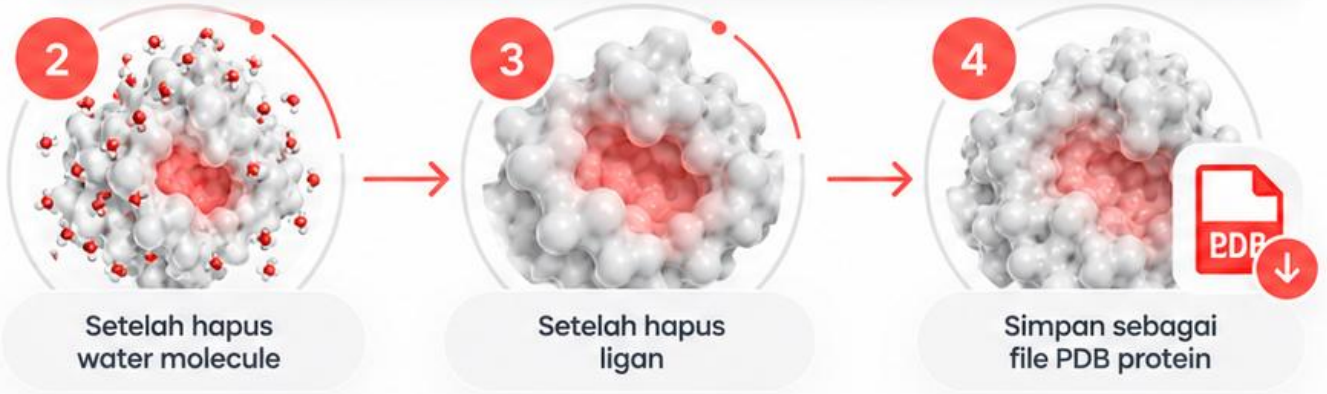
- posisi **ligan ko-kristal**
- **PrankWeb / CavityPlus**
- **residu aktif** dari literatur



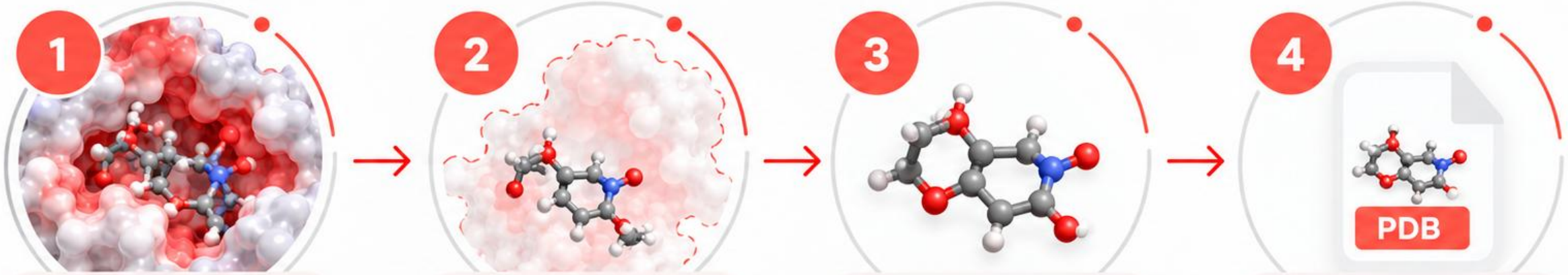
Preparasi Protein di Discovery Studio

1. **buka** file PDB
2. **hapus** water molecule
3. **hapus ligan** untuk menyimpan reseptor
4. **simpan** sebagai file PDB protein





Preparasi Ligan Ko-kristal

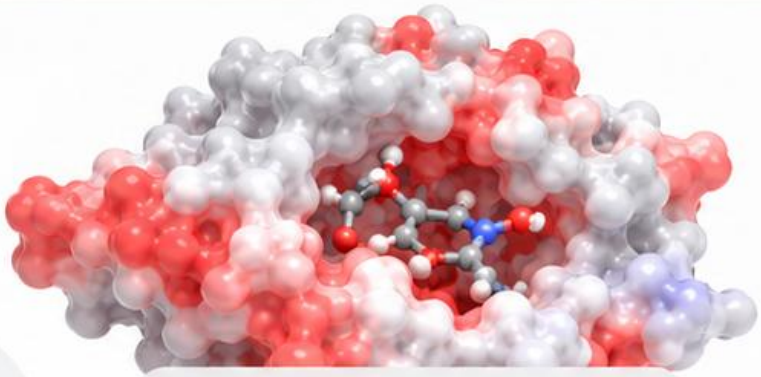


buka struktur kompleks awal

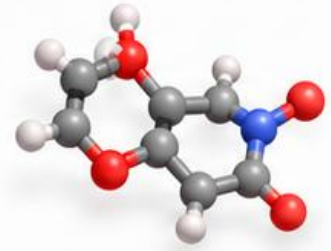
hapus protein

sisakan ligan ko-kristal

simpan sebagai PDB ligan



Kompleks protein–ligan (awal)



Ligan ko-kristal (hasil akhir)

Tujuan:
memisahkan native ligand dari kompleks

Catatan Penting Preparasi

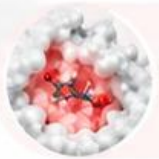
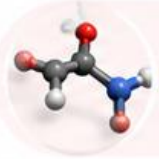
1. jangan salah menghapus **ligan kontrol**
2. pastikan ligan dan protein **tersimpan terpisah**
3. **nama file** harus rapi
4. buat folder: **receptor**, **native ligand**, **candidate ligand**, **output**

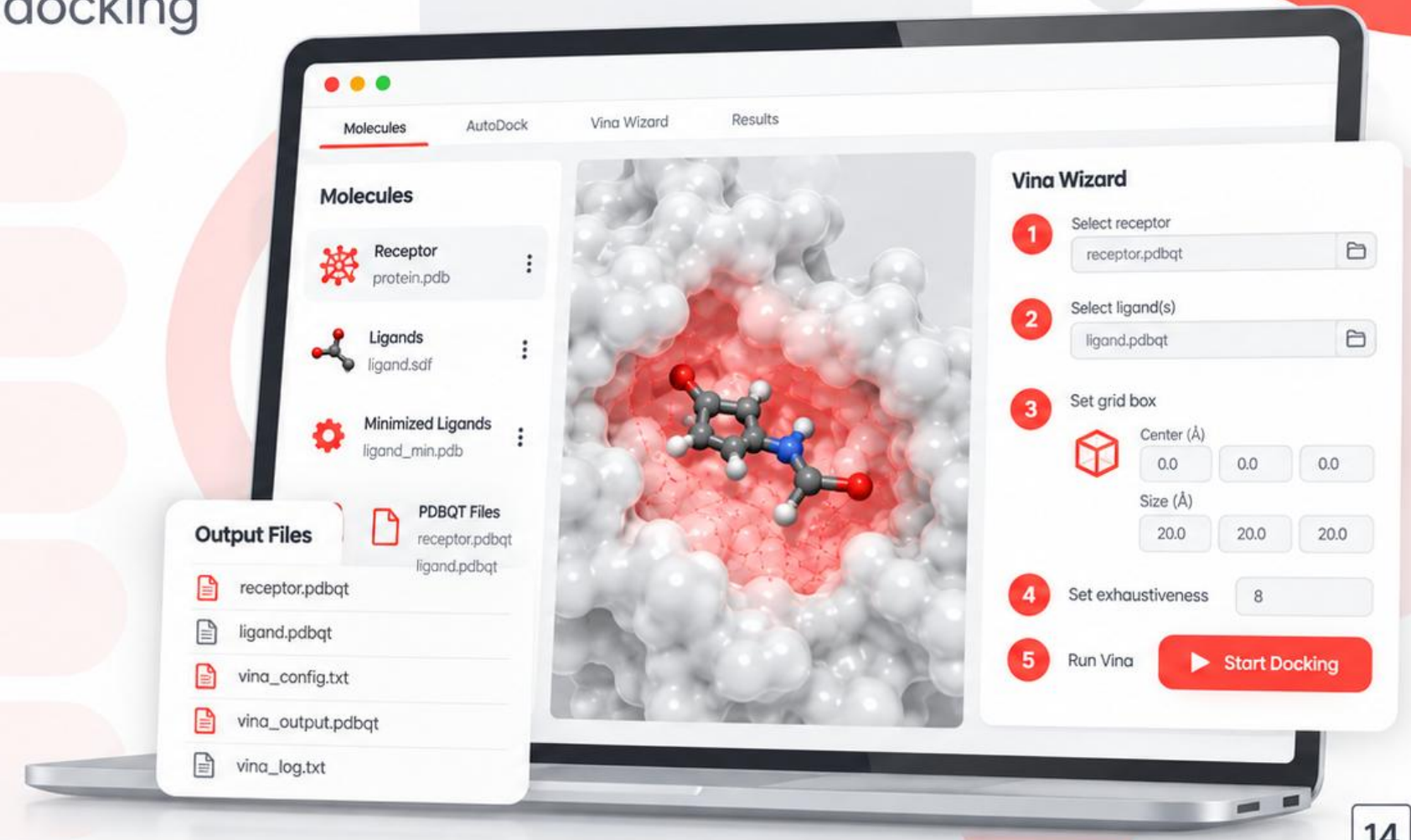


Hindari kesalahan sejak awal

PyRx dan AutoDock Vina

Preparasi file sebelum docking

1.  receptor sebagai **macromolecule**
2.  ligan sebagai **ligand**
3. lakukan **minimization**
4. convert ke **PDBQT**
5. lanjut ke **Vina Wizard**



Exhaustiveness

Exhaustiveness = intensitas pencarian pose.

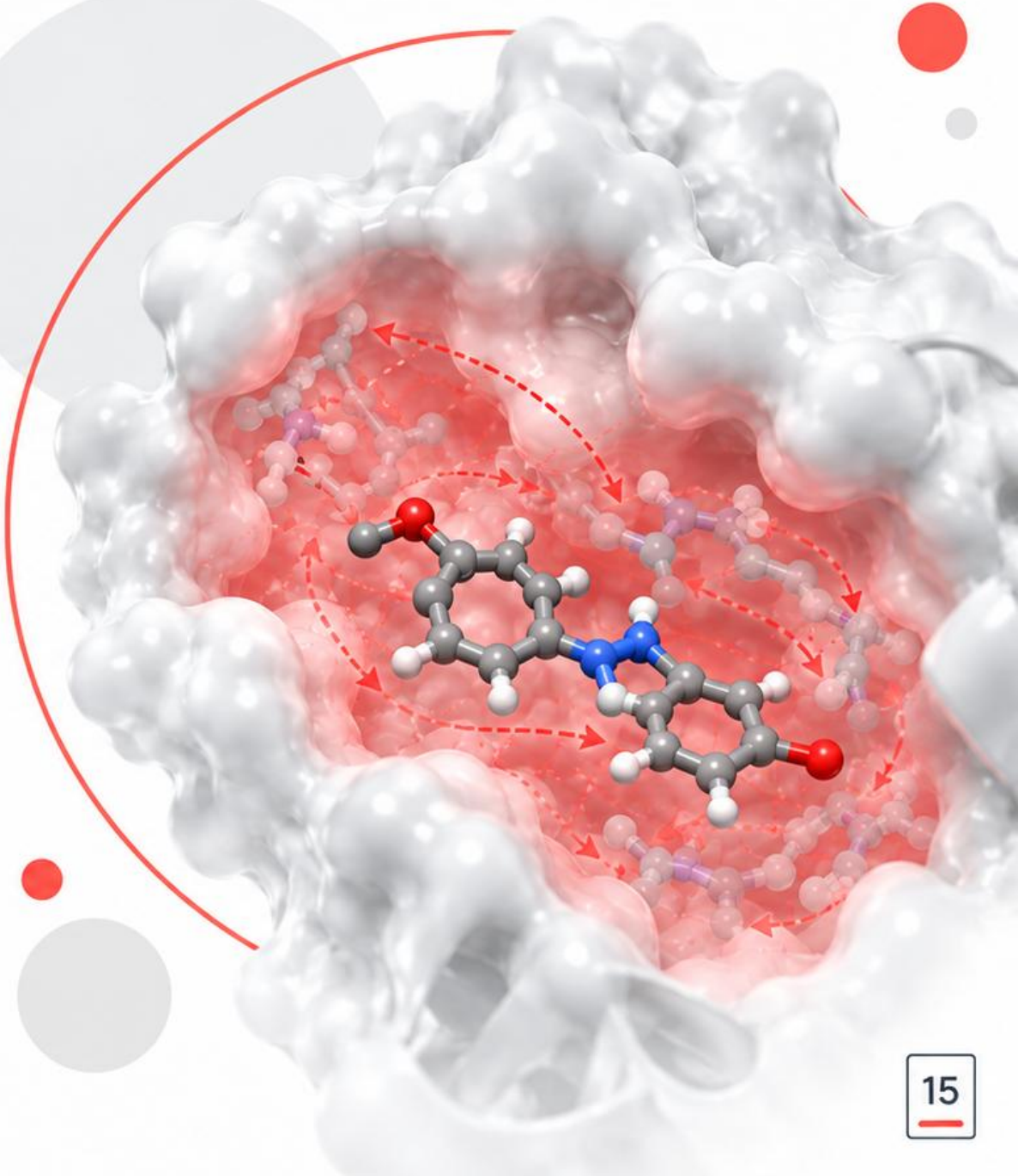
Nilai lebih tinggi → pencarian **lebih konsisten**

Tetapi komputasi **lebih lama**

Pencarian lebih cepat — Pencarian lebih konsisten

8 — 25 — 50 — 100

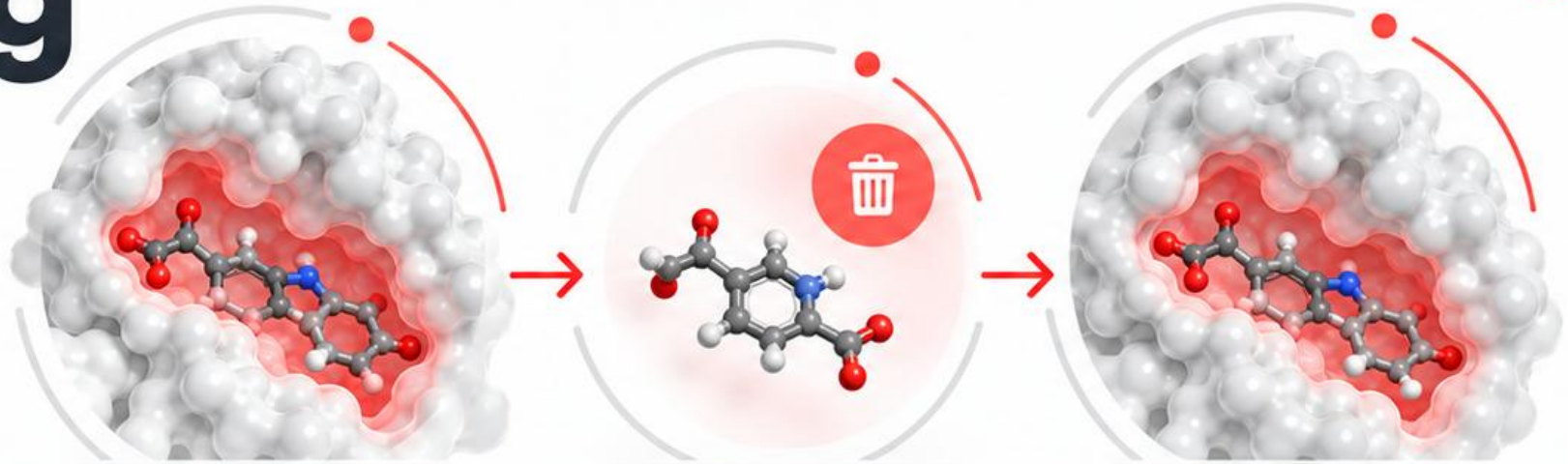
Digunakan saat menjalankan **Vina.**



Redocking

validasi metode

Redocking = ligan asli dilepas, lalu didocking ulang ke binding site yang sama.



Kompleks awal
(protein + ligan asli)

Ligan asli dilepas
dari binding site

Ligan didocking ulang
ke binding site yang sama

Tujuan Redocking

01 mengecek apakah metode mampu **mereproduksi pose eksperimental**

02 **bukan** membuktikan obat efektif

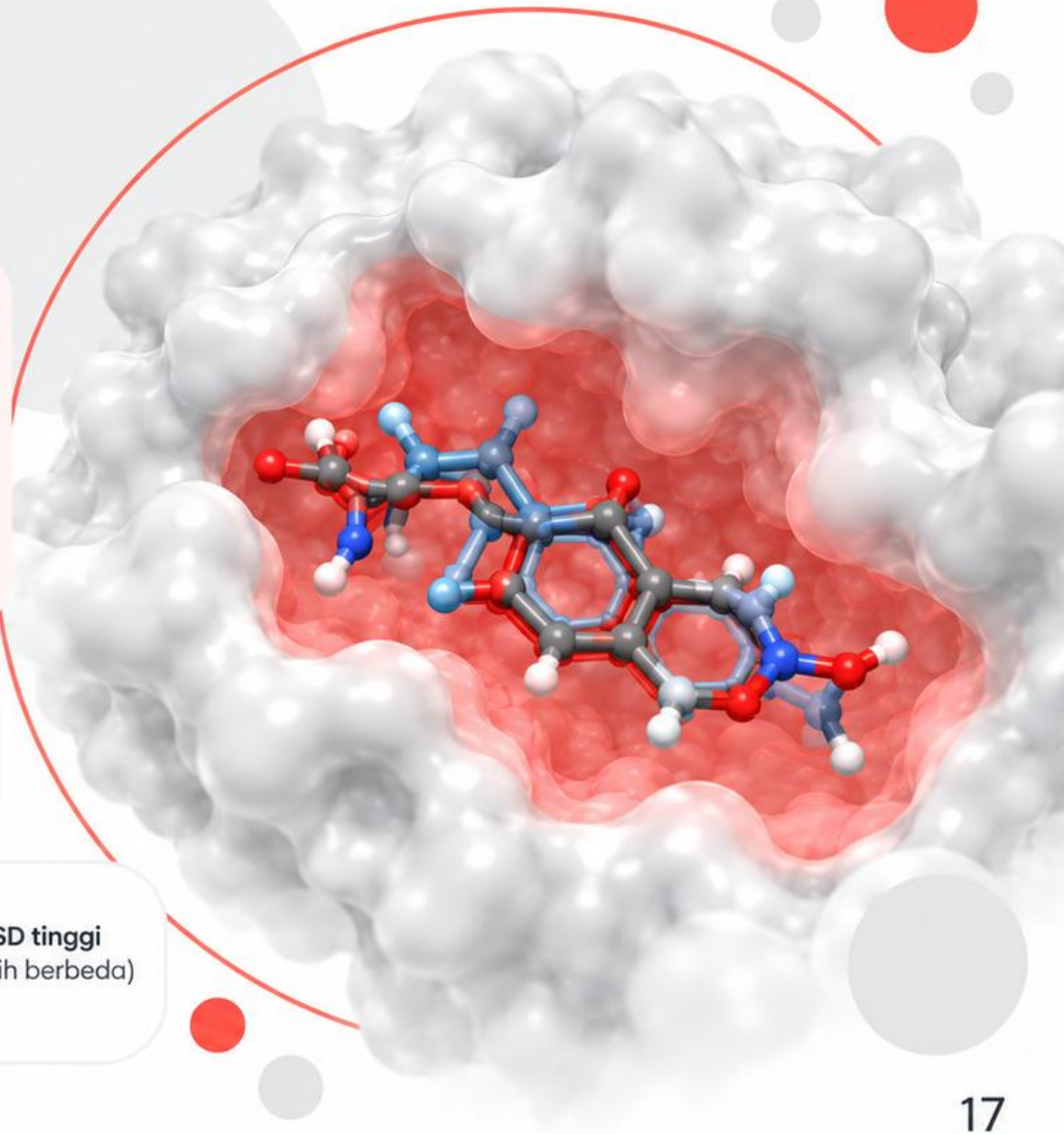
03 hanya **validasi internal workflow**

Interpretasi RMSD

RMSD ≤ 2 Å

Umumnya, redocking dianggap baik bila RMSD pose hasil docking terhadap pose kristal **≤ 2 Å**.

Tetapi ini bukan aturan absolut; interpretasi tetap perlu melihat plausibilitas interaksi.



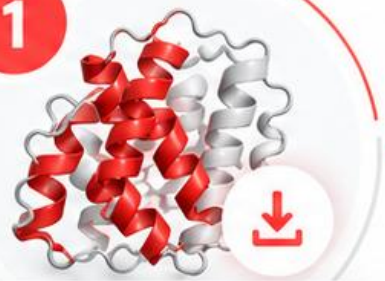


Source: PMC3823351

Praktik Mahasiswa

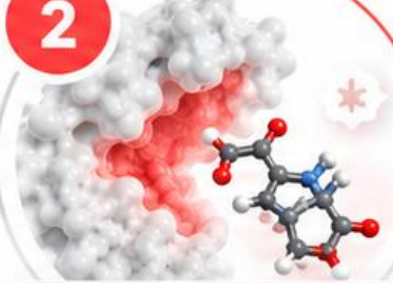
Tugas praktik pertemuan 1

1



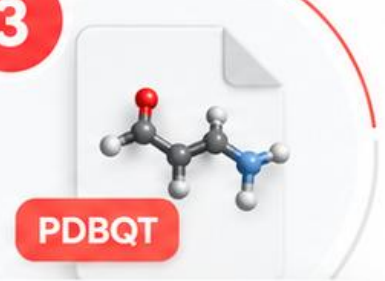
download protein

2



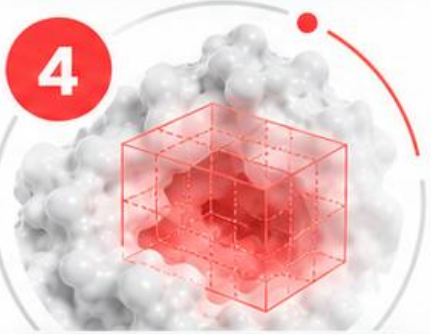
pisahkan protein dan ligan

3



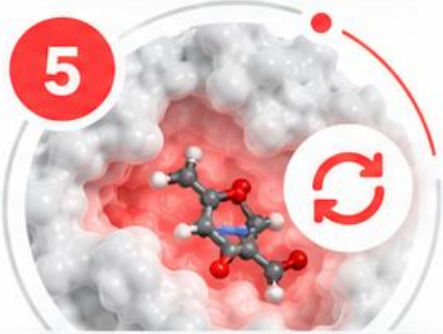
siapkan file **PDBQT**

4



buat **grid box**

5

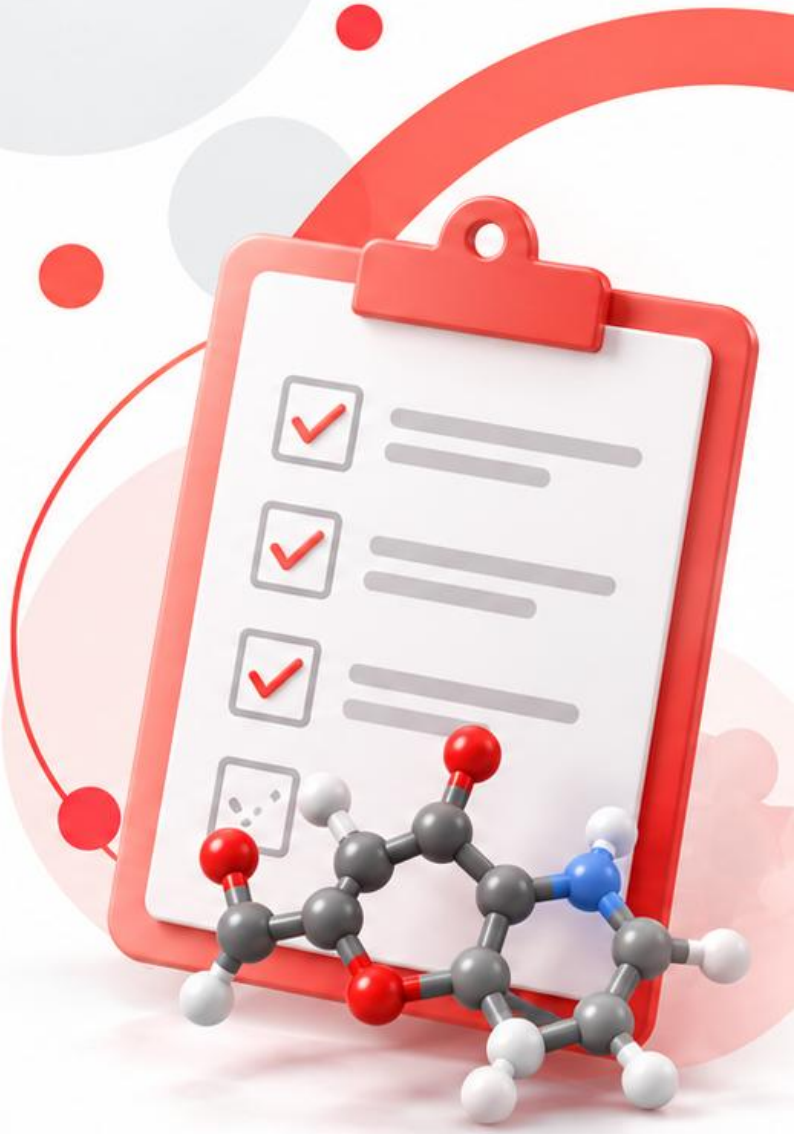


jalankan **redocking**

6



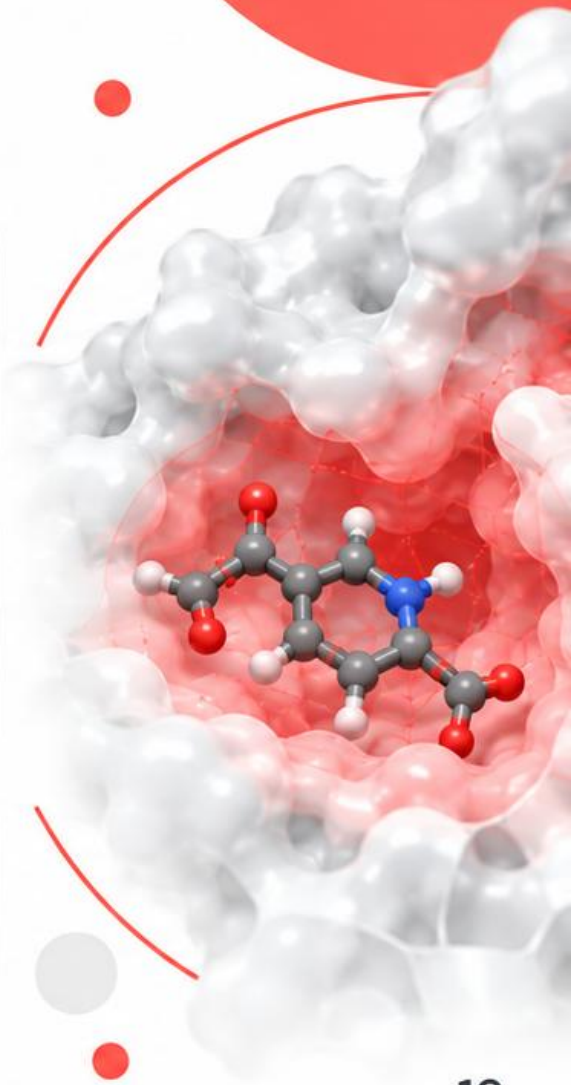
catat **affinity** dan **RMSD**



Output Pertemuan 1

Mahasiswa mengumpulkan

- file receptor
- file native ligand
- screenshot grid box
- tabel affinity redocking
- nilai RMSD
- kesimpulan: valid / tidak valid



Fokus pada kerapian file dan interpretasi hasill.

Penutup Pertemuan 1

Pesan utama

> **Docking yang baik dimulai dari preparasi dan validasi, bukan dari nilai *affinity* semata.**

